11 Big data visualization

This chapter covers

- Creating large random datasets of multiple types
- Using HTML5 canvas in conjunction with SVG to draw large datasets
- Optimizing geospatial, network, and traditional dataviz
- Working with quadtrees to enhance spatial search performance

This chapter focuses on techniques to create data visualization with large amounts of data. Because it would be impractical to include a few large datasets, we'll also touch on how to create large amounts of sample data to test your code with. You'll use several layouts that you saw earlier, such as the force-directed network layout from chapter 6 and the geospatial map from chapter 7, as well as the brush component from chapter 9, except this time you'll use it to select regions across the x- and y-axes.

This chapter touches on an exotic piece of functionality in D3: the quadtree (shown in figure 11.1). This is an advanced technique we'll use to improve interactivity